## a Validation targets support different claims

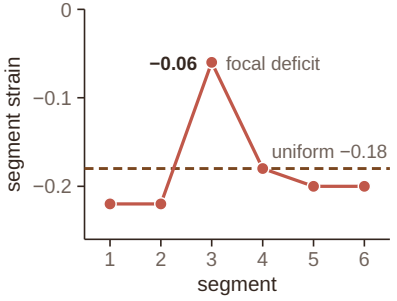
| ● Validation target | ✓ Direct claim it supports | ✕ Does not establish on its own |
|---|---|---|
| **Known motion**<br>phantom, STRAUS, MRXCAT2.0 | technical error in the simulated regime; capacity needs direct field fitting | in-vivo accuracy |
| **Paired DENSE or tagging**<br>same session, registered | material-sensitive agreement under matched definitions | focal location or extent, unless measured |
| **Scan–rescan**<br>same subject, re-imaged | precision; accuracy needs a reference | bias |
| **LGE or tissue characterisation**<br>scar or fibrosis label | biological association; motion error stays unmeasured | pointwise displacement error |
| **Outcome**<br>events, prognosis | prognostic association; utility needs a decision study | decision impact |

## b Proposed programme: combine complementary tests

| Stage | What is varied or fixed | Resources (Table 4) |
|---|---|---|
| **Pre-specify**<br>measurement contract | estimand, reference, tolerance and data splits fixed in advance | |
| **Stage 1**<br>Known motion | vary width, extent, amplitude, timing, resolution and noise | STRAUS **3 templates × 6 states**<br>MRXCAT2.0 **programmable generator**<br>CMAC **dynamic phantom** |
| **Stage 2**<br>Paired material reference | cine with DENSE or tagging in one session; registered; reference blinded | Stanford **55; pairs unverified**<br>CMAC **15 volunteers, 12 landmarks** |
| **Stage 3**<br>Tissue and decisions | LGE, responsiveness, decision impact | Clinical cohorts **LGE or outcomes** |

Counts are resource units, not independent test participants.

## c Same mean, focal deficit




| | |
|---|---|
| segments 1–3, focal | **−0.22 · −0.22 · −0.06** |
| segments 4–6, focal | **−0.18 · −0.20 · −0.20** |
| global mean, both fields | **−0.18** |
| range, focal / uniform | **0.16 / 0** |

Recovering the mean does not establish deficit recovery.